Question 6

- Suppose that the signal $x(t) = e^{j\omega t}$ is applied as an input to an LTI system with impulse response $h(t)$, show that the resulting output is $H(j\omega)e^{j\omega t}$ where $H(j\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$.
- Assume that the system is characterised by a 1st order D.E.:

$$\frac{dy(t)}{dt} + ay(t) = x(t)$$

If $x(t) = e^{j\omega t}$ for all t, then $y(t) = H(j\omega)e^{j\omega t}$ for all t; by substituting into the differential equation, determine $H(j\omega)$

Q6

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  Show that the resulting output is $H(j\omega)e^{j\omega t}$ where $H(j\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$.
- Assume that the system is characterised by a 1st order D.E.: $\frac{dy(t)}{dt} + ay(t) = x(t)$
  If $x(t) = e^{j\omega t}$ for all t, then $y(t) = H(j\omega)e^{j\omega t}$ for all t; determine $H(j\omega)$

a) $$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau = \int_{-\infty}^{\infty} h(\tau)e^{j\omega(t-\tau)}d\tau = e^{j\omega t}\int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau = e^{j\omega t}H(j\omega)$$

b) $y(t) = e^{j\omega t}H(j\omega)$

$$\frac{dy(t)}{dt} = j\omega e^{j\omega t}H(j\omega)$$

Sub into D.E.:

$$j\omega e^{j\omega t}H(j\omega) + ae^{j\omega t}H(j\omega) = e^{j\omega t}$$

$$j\omega H(j\omega) + aH(j\omega) = 1$$

$$H(j\omega) = \frac{1}{a + j\omega}$$